HYDROSEISMIC

Earthquake and Flood Incident Monitoring Prototype

A Python and Jupyter project for combining public incident data, validating records, visualizing locations, and exploring transparent prototype priority scores.

1. Problem Statement

Earthquake and flood information is published by multiple sources. Different formats make it harder to explore incidents together.

Goal: create one normalized dataset and an interactive way to filter, inspect, and visualize selected incidents.

2. Data Sources

USGS: earthquake event records.

GDACS: earthquake and flood event records, with alert levels where available.

Current dataset snapshot: 288 records.

USGS: 188 | GDACS: 100

3. System Workflow

Public source feeds

↓

Retrieve and normalize data

↓

Combine incident records

↓

Validate fields and timestamps

↓

Calculate prototype priority scores

↓

Explore filters, tables, plots, and reports

4. Technology Stack

Python: project logic and processing

Pandas: data cleaning and tabular analysis

Jupyter Notebook: interactive development and demonstration

Matplotlib: charts and location plots

ipywidgets: interactive filtering controls

5. Dataset Snapshot

Total incidents: 288

Earthquakes: 277

Floods: 11

Calculated priority bands:

High: 5

Medium: 99

Low: 184

6. Prototype Priority Calculation

Earthquake magnitude contribution: up to 70 points.

GDACS alert contribution: Red 30, Orange 20, Green 10.

Missing alert: 0 points. Total score capped at 100.

Low: below 30 | Medium: 30 to below 60 | High: 60 to 100.

This is an experimental heuristic, not a validated risk model.

7. Data Quality

288 records checked.

Duplicate incident IDs: 0

Duplicate complete rows: 0

Missing coordinates: 0

Invalid latitude/longitude values: 0

Invalid timestamps: 0

These checks assess data structure, not completeness or impact accuracy.

8. Demonstration Plan

1. Load the scored master dataset.
2. Show the source and event-type counts.
3. Filter to USGS and then GDACS.
4. Filter to earthquakes and then floods.
5. Select the High priority band.
6. Show the location plot and priority chart.
7. Finish with the data-quality report.

9. Limitations and Future Work

Calculated priorities are not official disaster warnings.
Scores do not predict future events, damage, or casualties.
Data counts can change when source feeds refresh.

Future work: automate refreshes, add a basemap, improve tests, document data provenance, and evaluate scoring against independently verified impact data.

10. Conclusion

HydroSeismic demonstrates public-data integration, normalization, validation, visualization, and filtering.

The current 288-record dataset passed the reported readiness checks.

The next stage is improving repeatability and evaluating the prototype before considering operational use.